

Appendix to the Manuscript Enabling Semi-Supervised Travel Mode Prediction Through Synthetic Unlabelled Trip Instances

1 Data used in the experimental part of the study

The survey data were collected by a marketing research company that followed quality control and data protection standards. The research team developed the questionnaires. The contractor was selected through a legally mandated public procurement process. All participants were briefed on the survey’s objectives and provided their informed consent for data collection by reading and signing the consent form.

All data sets used in this work rely on the surveys summarised in the main manuscript. Further details on the way the surveys providing data used in this work were performed can be found in [1].

2 Additional summary of the performance of the models

Figures 1, 2, 3, 4 and 5 present in the main manuscript document accuracy changes arising from the use of SU instances compared to the scenario of not using SU instances. As the aforementioned figures concern only some of the methods and settings we provide here figures relying on the same convention, but for all methods and all data sets.

3 Drift detection

Notably, the variable most frequently affected by drift was duration, with five distinct changes observed in the travel time if performed with public transportation. Conversely, the least frequent detection occurred for the number of transfers, where only a single instance of drift was recorded.

The drift detection analysis was conducted using the CW1 and CW2 datasets. The data in the CW2 dataset originates from a period following the collection of CW1. The results of the drift detection are presented in Tables 6, 7 and 8, with red vertical lines indicating the detected drift points, and a black vertical line marking the end of the CW1 dataset and the beginning of the streaming processing of the CW2 dataset.

References

- [1] Hassani, A., Nicińska, A., Drabicki, A., Zawojńska, E., Sousa Santos, G., Kula, G., Grythe, H., Zawieska, J., Jaczeńska, J., Rachubik, J., Archanowicz-Kudelska, K., Zagórska, K., Grzenda, M., Kubecka, M., Luckner, M., Jakubczyk, M., Wolański, M., Castell, N., Gora, P., Skedsmo, P.W., Rożynek, S., Horosiewicz, S.: Air quality and transport behaviour: sensors, field, and survey data from warsaw, poland. *Scientific Data* **11**(1), 1305 (Nov 2024). <https://doi.org/10.1038/s41597-024-04111-4>

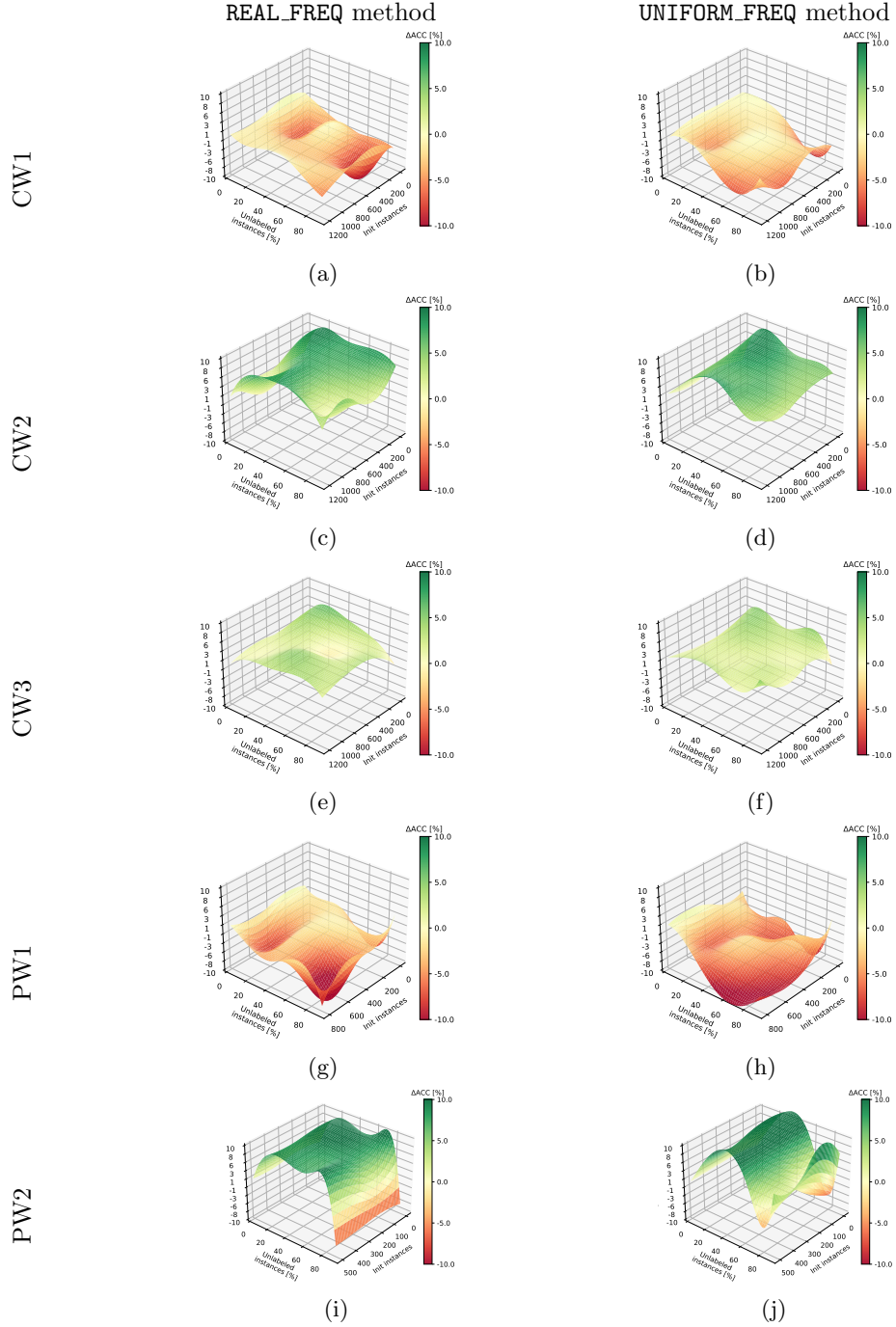


Figure 1: Accuracy change of cluster-and-label (CL) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q . Best viewed in colour.

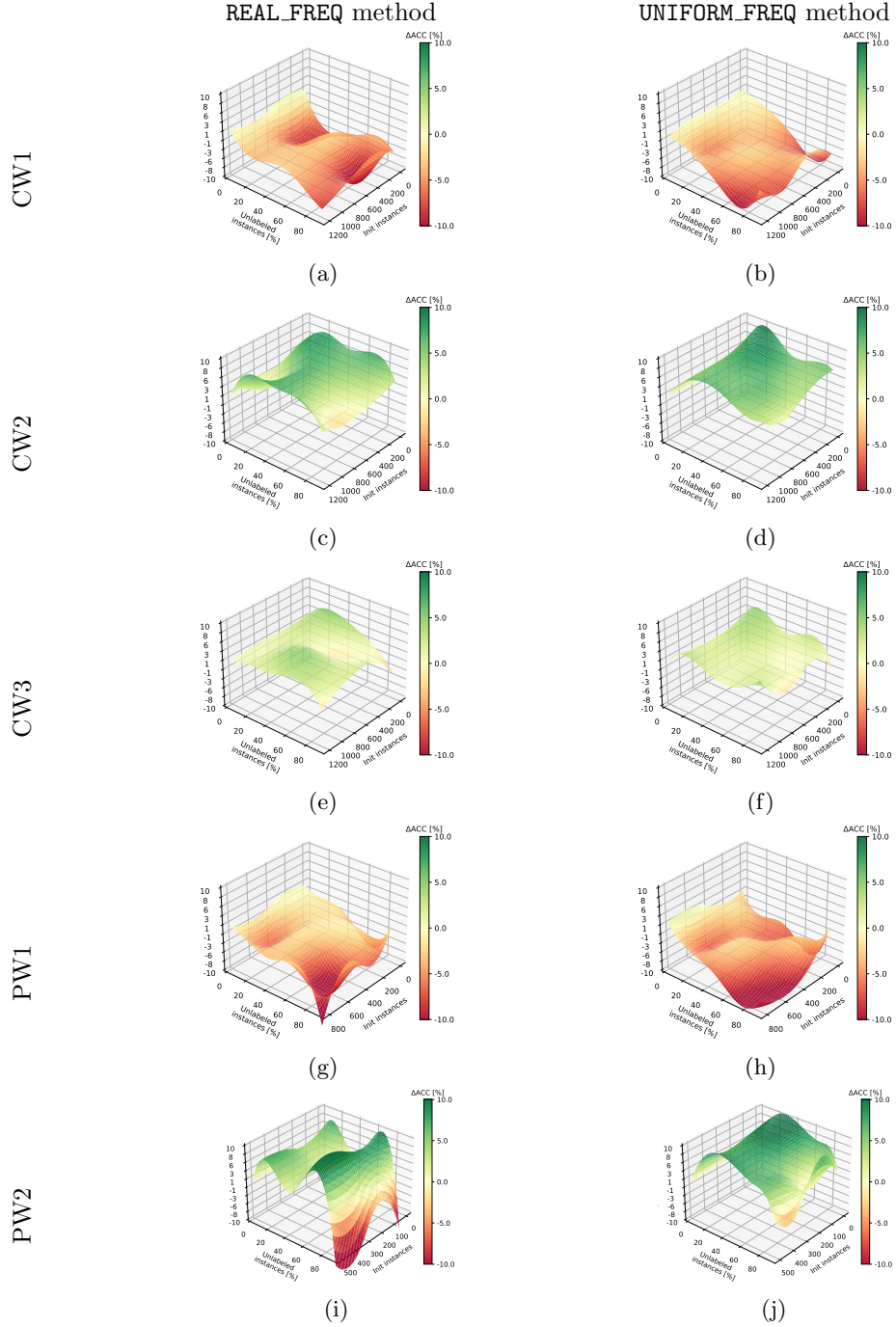


Figure 2: Accuracy change of cluster-and-label with pseudo labelling (CL^L) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q . Best viewed in colour.

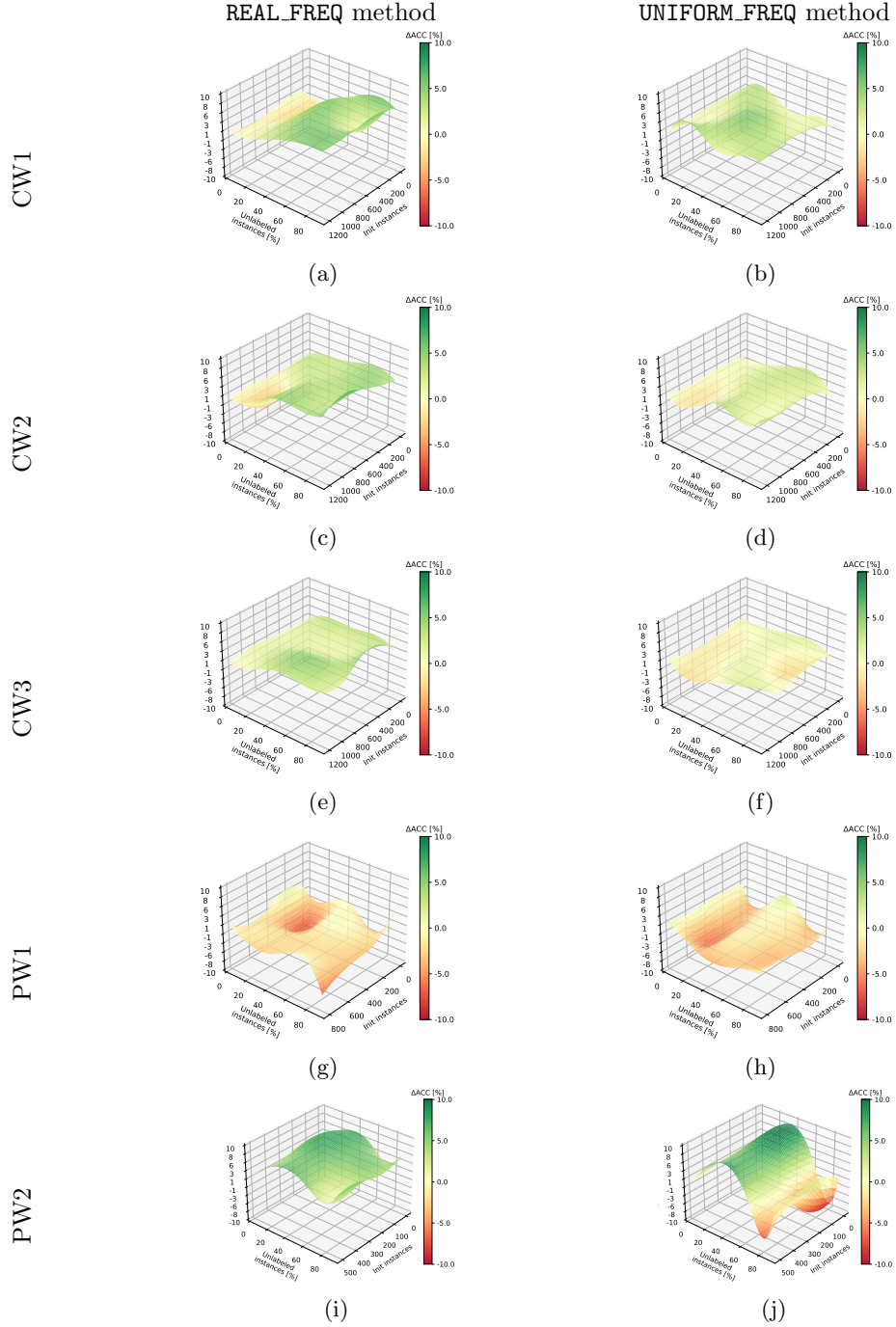


Figure 3: Accuracy change of cluster-and-label with sublearner (CL_{Sub}) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q . Best viewed in colour.

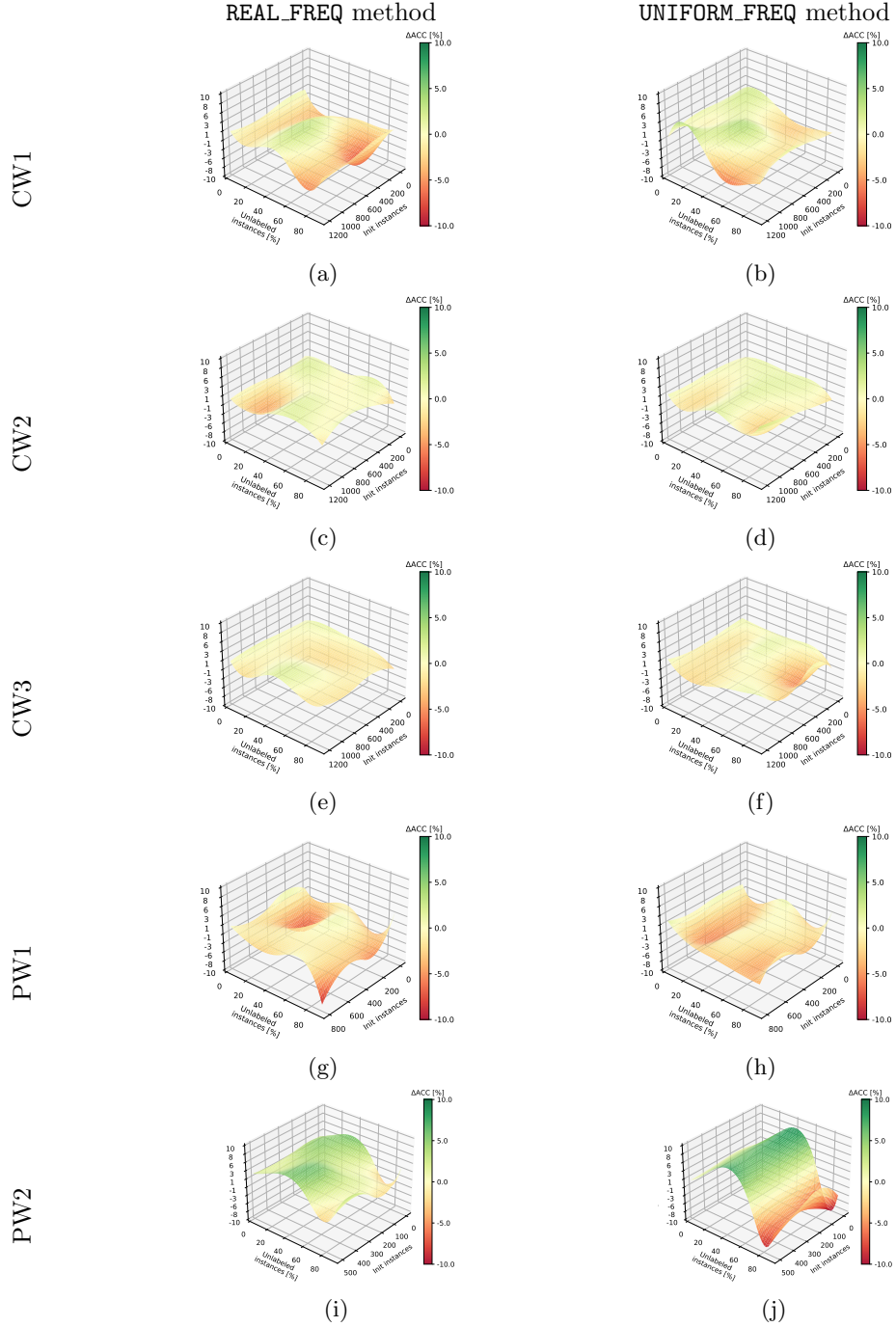


Figure 4: Accuracy change of cluster-and-label with sublearner and pseudo labelling ($\text{CL}_{\text{Sub}}^{\text{L}}$) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q . Best viewed in colour.

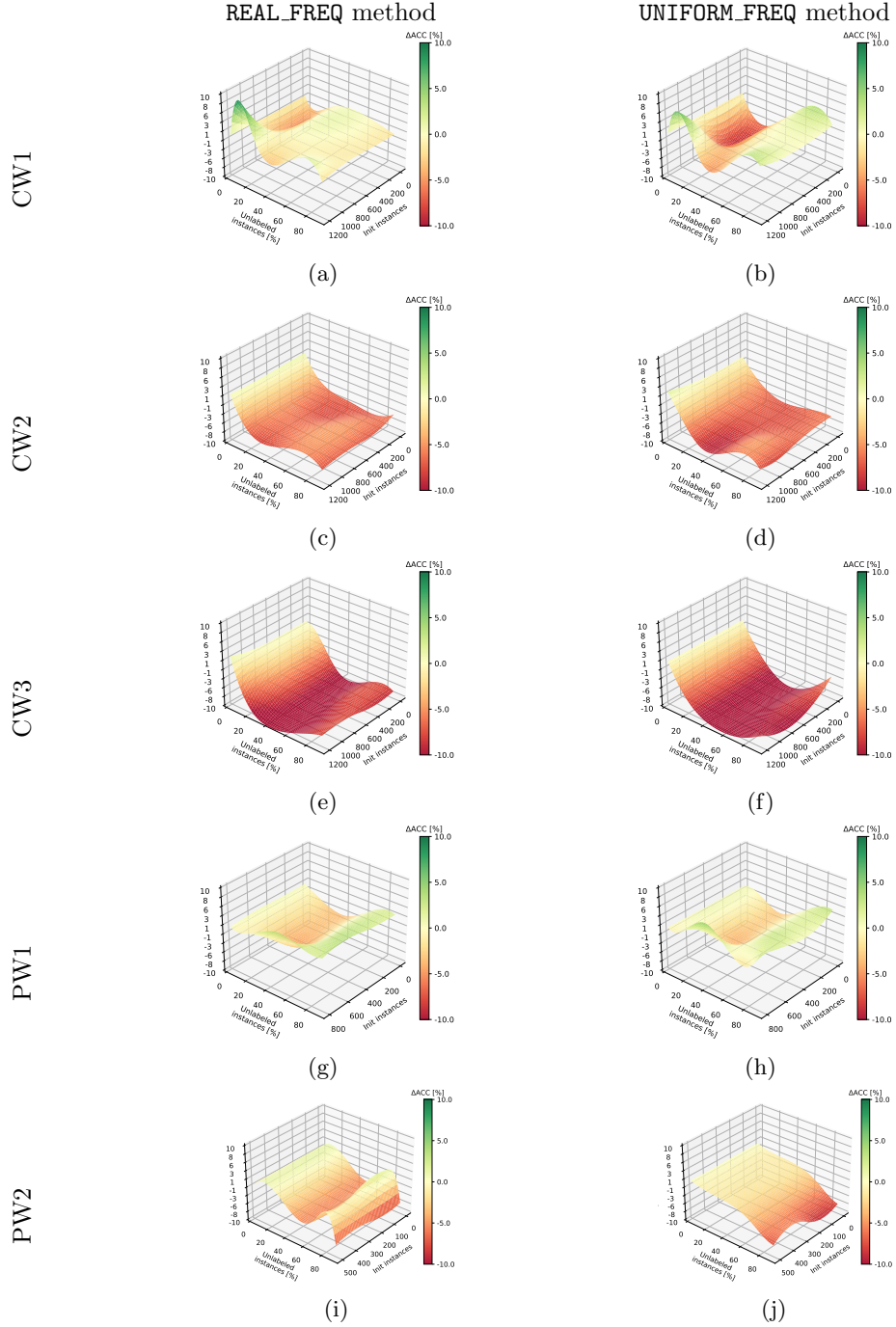
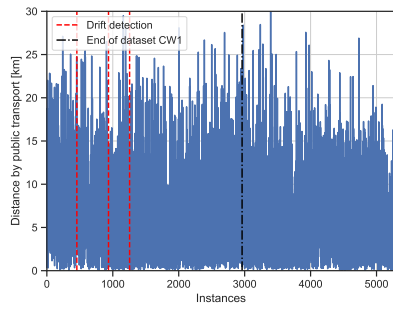
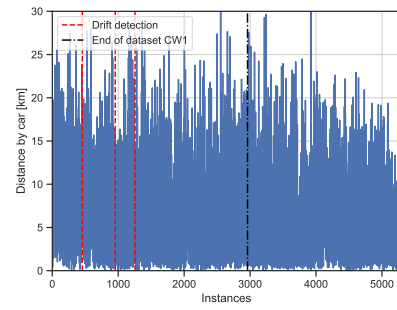


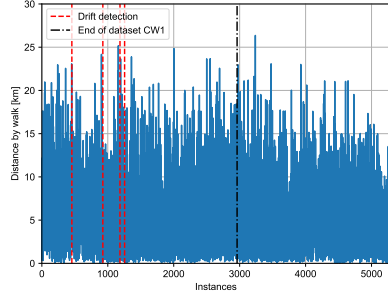
Figure 5: Accuracy change of self-training (SELF) predictions as a function of the percentage of unlabelled instances p and the number of initially used labelled instances q . Best viewed in colour.



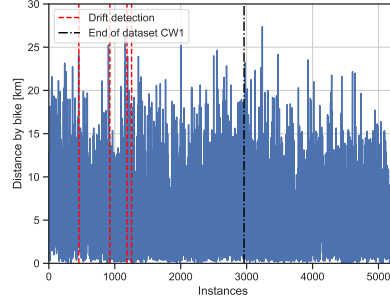
(a) Public Transport Distance



(b) Car Distance

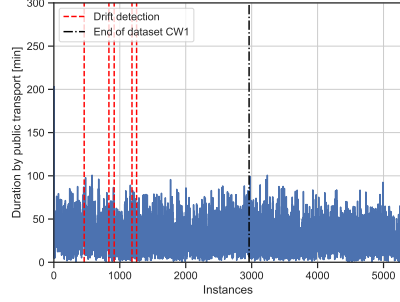


(c) Walk Distance

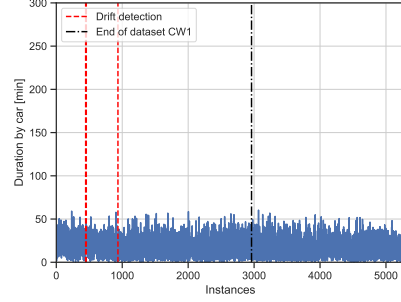


(d) Bike Distance

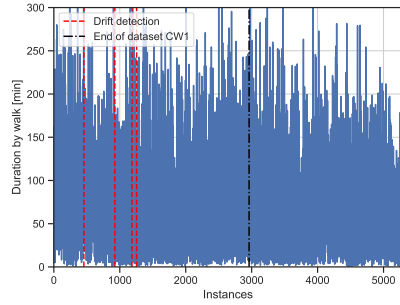
Figure 6: The travel distances recorded in successive surveys over time, with the points of detected drift indicated.



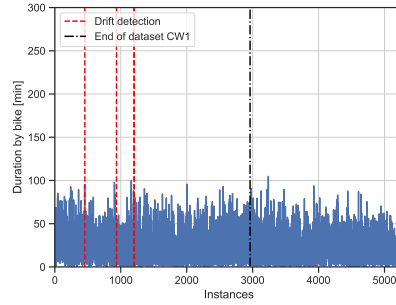
(a) Public Transport Duration



(b) Car Duration

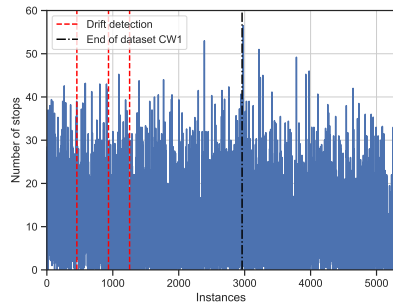


(c) Walk Duration

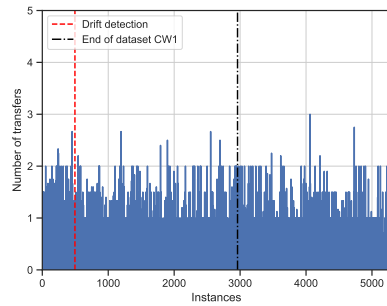


(d) Bike Duration

Figure 7: The travel duration recorded in successive surveys over time, with the points of detected drift indicated.



(a) Number of stops



(b) Transfers number

Figure 8: The number of stops (Fig. 8a) and the recorded instances of transfer (Fig. 8b), as documented in successive surveys over time, with the points of detected drift indicated.